

AURA

ESP32-Based Desk Companion Robot

Product Design Report

Revision 1.0 — June 2026



Aura prototype on a desk — emotive OLED eye display active

1. Project Overview

Aura is a compact, cube-shaped desk companion robot built around the ESP32 microcontroller. Its primary function is to display simple emotive expressions — currently a pair of glowing "eyes" rendered on an OLED display — to give the device a friendly, ambient personality on a desk or workspace. The enclosure is a small 3D-printed cube finished in a coral/pink shell, with a recessed front window through which the eye display is visible.

This report documents the physical design of the Aura enclosure from four standard orthographic views (front, back, left, right), along with the internal ESP32-based electronics layout.

1.1 Design Goals

- Small footprint suitable for permanent placement on a desk (approx. 80 x 60 x 50 mm)
- Friendly, toy-like aesthetic using a single bold accent color
- Clear visual focal point — the OLED "eyes" — centered on the front face
- Simple, low-cost electronics built around a single ESP32 dev board
- USB-C powered, cable routed out through the rear face
- Enclosure shell with open/vented top and side cut-outs for assembly access and heat dissipation

2. System Specifications

Device Name	Aura — ESP32 Desk Companion Robot
Microcontroller	ESP32 (Dual-core, Wi-Fi + Bluetooth, 38-pin Dev Board)
Display	0.91" / 0.96" OLED display (I2C, SSD1306) — renders animated eye expressions
Enclosure Material	3D-printed PLA, single-color (coral/pink) shell
Overall Dimensions	Approx. 80 mm (W) x 60 mm (D) x 50 mm (H)
Power Input	5V via USB-C / Micro-USB cable, routed through rear panel
Internal Wiring	Jumper wires connecting OLED to ESP32 I2C pins (SDA/SCL), routed inside cavity
Top Panel	Open-frame top for assembly access and component visibility
Side Cut-outs	Rectangular ventilation / cable-management slots on left and right faces
Mounting	Free-standing on flat desk surface, no external mounting required

3. Orthographic Views

The following sections present the Aura enclosure from the four primary orthographic angles. Reference photographs of the working prototype are included alongside descriptions of each face.

3.1 Front View



Figure 1 — Front view of Aura, showing the OLED eye display

The front face is the primary interaction surface. A rectangular recessed window is centered horizontally, behind which the OLED display sits flush with the inner wall of the enclosure. When powered on, the display renders two glowing white "eye" shapes against a black background, giving Aura its character. The surrounding shell is a flat coral/pink panel with no additional buttons or ports on this face.

3.2 Back View



Figure 2 — Rear/internal view of Aura, showing wiring and power cable exit

The rear face houses the power entry point. A USB cable exits through a cut-out at the base of the back panel, supplying 5V to the ESP32 board housed inside the cavity. The top edge of the

enclosure is open, exposing the internal jumper wires that connect the OLED display to the ESP32's I2C pins (SDA, SCL, VCC, GND). This open-top design allows easy access to the ESP32 board for programming via USB and for future modification.

3.3 Left Side View

The left side panel is a flat coral/pink face featuring a rectangular cut-out near the lower-front edge. This opening serves as a cable-routing channel and ventilation slot, allowing airflow around the ESP32 board and providing a pass-through for any additional sensor or peripheral wiring. The panel height matches the overall enclosure height of approximately 50 mm, with the cut-out positioned roughly one-third of the way up from the base.

3.4 Right Side View

The right side panel mirrors the left side, featuring a matching rectangular cut-out for symmetry and additional ventilation. Together, the left and right cut-outs allow heat generated by the ESP32 and OLED driver circuitry to escape the enclosure, preventing the compact cavity from overheating during continuous operation. Both side panels are otherwise plain, continuing the flat coral/pink finish of the front and back faces.

3.5 Top View

The top of the enclosure is intentionally left open in the current prototype, exposing the ESP32 board and internal wiring loom. This open-frame top doubles as an assembly access point and a visual "window" into the robot's electronics, reinforcing the maker-style aesthetic. In a production version this could be closed with a snap-fit lid while retaining a small slot for the USB programming port.

4. Electronics & Wiring

4.1 Core Components

- ESP32 Dev Board — main controller, handles display driving logic and (optionally) Wi-Fi connectivity
- OLED Display (SSD1306, I2C) — mounted behind the front window, displays animated eye graphics
- USB-C / Micro-USB cable — power and programming connection, routed out the rear panel
- Jumper wires — connect OLED VCC, GND, SDA, SCL to corresponding ESP32 GPIO pins

4.2 Connection Table

OLED VCC	ESP32 3V3
OLED GND	ESP32 GND
OLED SDA	ESP32 GPIO 21 (I2C Data)
OLED SCL	ESP32 GPIO 22 (I2C Clock)
Power In	ESP32 5V / VIN via USB cable

4.3 Operating Notes

- The OLED communicates with the ESP32 over the I2C bus at default address 0x3C
- Eye animations are driven by firmware running on the ESP32, allowing for blinking, looking-around, and mood-based expressions
- The open enclosure design simplifies re-flashing firmware over USB without disassembly

5. Summary

Aura is a compact, friendly desk companion built on a simple but effective design: a coral 3D-printed cube housing an ESP32 microcontroller and an OLED display that renders animated eyes through a front-facing window. The open top and side ventilation slots keep the design accessible for prototyping while the flat, single-color shell gives it a clean, toy-like presence on any desk. This design is ready for refinement into a fully enclosed version with a snap-fit lid, integrated speaker, or additional sensors for future iterations.

— *End of Report* —